Skills Summary

Solving Logarithmic Equations with Domain Checks

Use this reference while completing your worksheet or when studying.

The domain rule (use it first, every time): $\log_b A$ exists only when $A > 0$ (and $b > 0$, $b \neq 1$). Write the restriction on x *before* solving; clearing logarithms can invent candidates the original equation never had.

Conversion: $\log_b A = c \iff A = b^c$. **Properties (valid where every argument is positive):** $\log_b M + \log_b N = \log_b(MN)$; $\log_b M - \log_b N = \log_b \frac{M}{N}$; $p \log_b M = \log_b M^p$.

1. One log = a number (log-to-exponential)

Isolate the single logarithm, then convert: $\log_b(\text{argument}) = c$ becomes argument $= b^c$. Solve the resulting equation and confirm the answer keeps the argument positive.

Example: Solve $\log_3(x + 2) = 4$.
Step 1. One logarithm, alone, equal to a number — that is the conversion case, so switch to exponential form.
Step 2. Domain $x + 2 > 0 \Rightarrow x > -2$; $x + 2 = 3^4 = 81$, so $x = \mathbf{79}$, and $79 > -2$ ✓

Try it: Solve $\log_5 x = 3$ (state the domain too).

check: $x = \mathbf{125}$

2. Log = log, same base (equate-like-logs)

$\log_b M = \log_b N \iff M = N$, **provided** $M > 0$ and $N > 0$. Drop the logarithms, solve, then test each candidate in *both* original arguments.

Example: Solve $\log_7(3x - 2) = \log_7(x + 8)$.
Step 1. Same base on both sides and nothing else in the way, so the logarithms cancel and the arguments must match.
Step 2. Domain: $3x - 2 > 0$ and $x + 8 > 0 \Rightarrow x > \frac{2}{3}$; $3x - 2 = x + 8$ gives $x = \mathbf{5}$, and both arguments equal $13 > 0$ ✓

Try it: Solve $\log_2(4x + 1) = \log_2(x + 10)$.

check: $x = \mathbf{3}$

3. Two logs on one side (combine-logs-then-solve)

Use the product, quotient, or power rule to make it *one* logarithm, then convert. This is where extraneous candidates appear: the quadratic you get comes from the equation *after* the logarithms are gone, so every root must be tested against the domain.

Example: Solve $\log_3 x + \log_3(x-8) = 2$.
Step 1. Two logarithms of the same base added on one side, so the product rule collapses them into one before converting.
Step 2. Domain $x > 8$; $x(x-8) = 3^2 = 9 \Rightarrow x^2 - 8x - 9 = 0 \Rightarrow (x-9)(x+1) = 0$; $x = -1$ fails $x > 8$, so $x = \mathbf{9}$

Try it: Solve $\log_2(x+1) + \log_2(x-1) = 3$.

check: $\boldsymbol{x = 3}$

4. Decimal answers (approximate-log-solutions)

Convert to exponential form, keep the exact expression ($e^2 - 3$, $2^{4.7} - 5$) through the algebra, and round only on the last line. The exponent need not be a whole number. Then check the rounded value against the domain.

Example: Solve $\ln(x+3) = 2$ to the nearest hundredth.
Step 1. A single natural logarithm equals a number, so convert with base e and round only at the end.
Step 2. Domain $x > -3$; $x + 3 = e^2$, so $x = e^2 - 3 = 7.389\ldots - 3 \approx \mathbf{4.39}$, which satisfies $x > -3$ ✓

Try it: Solve $\log_3(x+1) = 2.4$ to the nearest hundredth.

check: $\boldsymbol{x \approx 12.97}$